

$\pi_{0.5}$ · arXiv 2504.16054 · Physical Intelligence

Archaeologist

11+3 · seat 3

RT → OpenVLA → π_0 → FAST → $\pi_{0.5}$
→ KI → RTC → Hi Robot · BEHAVIOR

Sep 30 paper club · VLA Architectures

Spine: open-world + hybrid FAST discrete pretrain → FM retained

Semantic subtask → FM chunk (~50-step/~1s @ 50 Hz) · NOT a classical planner · RTC companion

Club spine · one-liner

Open-world VLA via heterogeneous co-training + hierarchical inference:

FAST discrete pretrain → FM retained for continuous control.

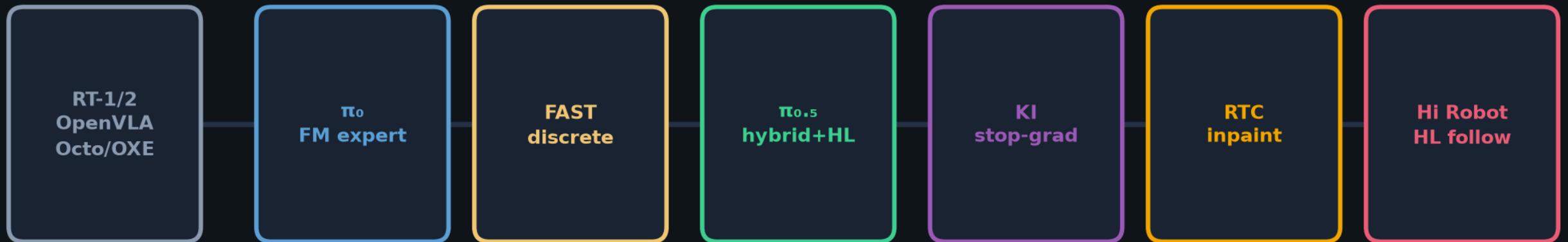
Semantic subtask → ~50-step / ~1s FM chunk @ 50 Hz. NOT a classical planner.

RTC companion for async soft-inpaint. Blogs: “FAST replaced FM” = FALSE.

BEHAVIOR'26: $\pi 0.5$ + GR00T baselines · Oct 2 = $\pi 0.5$ -baseline + deltas.

Lineage timeline · Archaeologist spine

RT/OpenVLA/Octo/OXE $\rightarrow$ π_0 $\rightarrow$ FAST $\rightarrow$ $\pi_{0.5}$ $\rightarrow$ KI $\rightarrow$ RTC $\rightarrow$ Hi Robot



Challenge archaeology

BEHAVIOR-1K

household sim foundation

BEHAVIOR'25

50 tasks · $\pi_{0.5}$ winner ~26% q

BEHAVIOR'26

100 tasks · $\pi_{0.5}$ + GR00T baselines

What's new vs reused

Reused from π_0 / priors

- PaliGemma / SigLIP + Gemma init
- ~300M action expert pattern
- Flow matching continuous chunks
- Cross-embodiment padding / OXE
- Action chunking @ 50 Hz PD
- Beta low- τ timestep prior

New in $\pi_{0.5}$

- Heterogeneous mixture curriculum
- Unified HL/LL in one model
- Two-stage FAST→FM specialization
- Verbal instruction (VI) coaching
- Home-scale open-world eval (~100 homes)
- Knowledge insulation path → KI paper

π_0 VS $\pi_{0.5}$ · what changed

π_0 (2410.24164)

- FM action expert throughout
- No FAST pre-phase
- No unified HL head in recipe
- Lab / matched scenes focus
- ~10k h cross-embodiment
- H=50 · ~10 Euler · 20-50 Hz
- PaliGemma + ~300M expert

$\pi_{0.5}$ (2504.16054)

- FAST pretrain → FM post-train
- FM RETAINED at inference
- Unified HL subtask → LL chunk
- Open-world new homes KPI
- ~400h MM / ~100 homes + mix
- H=50 @ 50 Hz · 10 FM steps
- Co-train: MM ME CE HL WD VI

Novelty = mixture curriculum + hierarchy + hybrid recipe + home-scale eval — not new FM math

FM vs FAST · say this aloud

Hybrid FAST discrete pretrain → FM retained for continuous control

Claim	Verdict
FAST replaced FM	MYTH
FAST pretrain + FM post-train	TRUE
FM head retained at inference	TRUE
OpenPI FM API for $\pi_{0.5}$	CLEAR
FAST is the deployed LL controller	FALSE

OpenPI ships FM head for $\pi_{0.5}$ · KI (2505.23705) formalizes stop-grad FAST+FM

Hybrid recipe · FAST discrete pretrain → FM retained

Bust the myth: FAST does NOT replace FM at deploy

① PRE-TRAIN · 280k steps

Discrete next-token VLA

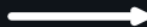
SigLIP 400M + Gemma ~2B

text · boxes · FAST action tokens · $\alpha=0$

FAST action tokens

scalable discrete pretrain signal

Mixture: MM · ME · CE · HL · WD



② POST-TRAIN / INFER · 80k

Joint NTP + flow matching $\alpha=10$

Action expert ~300M · FM

Ada-RMSNorm on τ · random init expert

Inference: AR subtask → 10 FM steps

~50-step / ~1s chunk @ 50 Hz PD

MM+ME+WD+VI · drop CE · FM RETAINED

NOT classical planner · same model for HL + LL · RTC = async soft-inpaint companion (2506.07339)

Co-training mixture · MM ME CE HL WD VI

97.6% of phase-1 examples are NOT target mobile household teleop

MM

~400h / ~100 homes

Target mobile manip

ME

Static multi-env

Cross-emb + diversity

CE

Lab + OXE

Task diversity

HL

Subtask (+boxes)

Hierarchy supervision

WD

Web VQA/caption

OOD semantics

VI

~11% of HL-MM

Verbal coaching

2026 BEHAVIOR map · CLEAR

Official baselines · scale · timeline · '25 preview

Official baselines

π 0.5 + GROOT N1.7

behavior.stanford.edu/challenge

Scale

100 tasks · 7 scenes · 20k demos

~1950 h · RGB+depth+proprio · BDDL

Timeline 2026

07/02



launch

10/16



submit

11/04



winners

'25 winner on π 0.5 · preview of '26 bottlenecks

RLC adapted π 0.5 → ~26% q (correlated FM noise · System-2 stages · multi-sample FM · compression)

Bottlenecks: precision grasps · non-Markov stages · recovery demos · Oct 2 → propose deltas on this substrate

Fig.3 · Two-stage: FAST pretrain → FM post-train

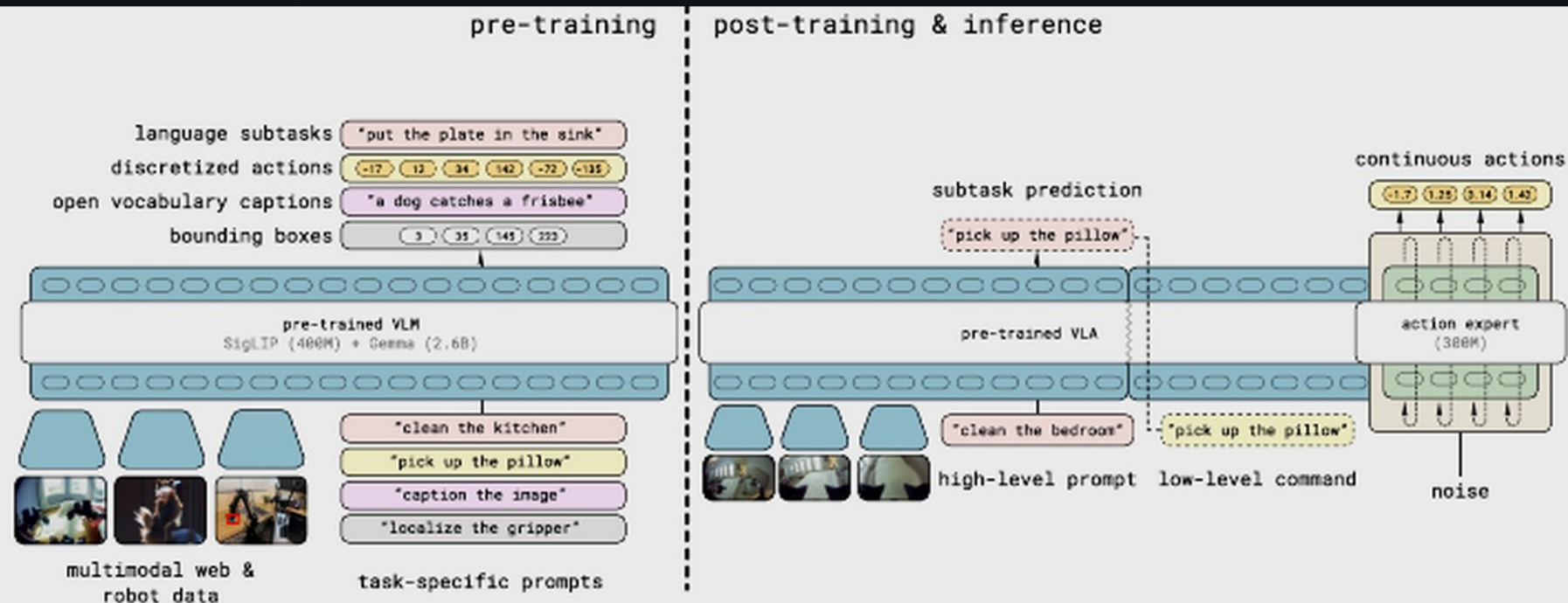


Fig. 3: **Model overview.** $\pi_{0.5}$ is trained in two stages. First, a pre-training stage combines all of the different data sources to produce an initial VLA with discrete tokens. This stage uses data from diverse robotic platforms, high-level semantic action prediction, and data from the web. Robotic data uses the FAST action tokenizer to represent actions as discrete tokens [64]. Second, a post-training stage specializes the model for low-level and high-level inferences for mobile manipulation, leveraging the most task-relevant data, including verbal instructions from human supervisors. This stage uses flow matching to represent the action distribution, enabling efficient real-time inference and the ability to represent fine-grained continuous action sequences. At inference time, the model first infers a high-level subtask, and then predicts the actions based on this subtask.

primitives, such as picking up objects, methods that allow for task-specific assumptions (e.g., grasp prediction, or incorporating model-based planning and control) have been shown to generalize broadly, even to entirely new homes [40, 20, 60, 56, 29]. However, such methods do not readily generalize to the full range of possible tasks that a generalist robot might need to perform. More recently, large-scale datasets collected across

input to output tokens. The weights of these models are initialized from pre-trained vision-language models. By encoding policy inputs and outputs into tokenized representations, the imitation learning problem described above can be cast as a simple next-token-prediction problem over a sequence of observation, instruction and action tokens, and we can leverage the scalable tools of modern machine learning to

$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

Physical Intelligence

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<https://pi.website/blog/pi05>



$$\pi_{\theta}(\mathbf{a}_{t:t+H}, \hat{\ell} | \mathbf{o}_t, \ell) = \pi_{\theta}(\mathbf{a}_{t:t+H} | \mathbf{o}_t, \hat{\ell}) \pi_{\theta}(\hat{\ell} | \mathbf{o}_t, \ell),$$

where the action distribution does not depend on ℓ , only on $\hat{\ell}$. Thus, high-level inference captures $\pi_{\theta}(\hat{\ell} | \mathbf{o}_t, \ell)$, and low-level inference captures $\pi_{\theta}(\mathbf{a}_{t:t+H} | \mathbf{o}_t, \hat{\ell})$, with both distributions represented by the same model.

The model corresponds to a transformer that takes in N multimodal input tokens $x_{1:N}$ (we use the term token loosely here, referring to both discretized and continuous inputs) and produces a sequence of multimodal outputs $y_{1:N}$, which we can write as $y_{1:N} = f(x_{1:N}, A(x_{1:N}), \rho(x_{1:N}))$. Each x_i can be a text token ($x_i^w \in \mathbb{N}$), an image patch ($x_i^I \in \mathbb{R}^{p \times p \times 3}$), or an intermediate denoising value of a robot action in flow matching ($x_i^a \in \mathbb{R}^d$). The observations $\mathbf{o}_t$ and ℓ form the prefix part of $x_{1:N}$. Depending on the token type, as indicated by $\rho(x_i)$, each token can be processed not only by a different encoder, but also by different expert weights within the transformer. For example, image patches are fed through a vision encoder, and text tokens are embedded with an embedding matrix. Following π_0 [8], we linearly project action tokens x_i^a into the transformer embedding space and use separate expert weights in the transformer to process the action tokens. The attention matrix $A(x_{1:N}) \in [0, 1]^{N \times N}$ indicates if a token can attend to another token. Compared to standard causal attention

tokenizer) and iterative integration of the flow field, combining the best of both worlds. We use the attention matrix to ensure that the different action representations do not attend to each other. Our model is optimized to minimize the combined loss

$$\mathbb{E}_{\mathcal{D}, \tau, \omega} \left[H(x_{1:M}, f_{\theta}^{\ell}(\mathbf{o}_t, \ell)) + \alpha \|\omega - \mathbf{a}_{t:t+H} - f_{\theta}^a(\mathbf{a}_{t:t+H}^{\tau, \omega}, \mathbf{o}_t, \ell)\|^2 \right], \quad (1)$$

where $H(x_{1:M}, y_{1:M}^{\ell})$ is the cross entropy loss between the text tokens and predicted logits (including the FAST encoded action tokens), $y_{1:H}^a = f_{\theta}^a(\mathbf{a}_{t:t+H}^{\tau, \omega}, \mathbf{o}_t, \ell)$ is the output from the (smaller) action expert, and $\alpha \in \mathbb{R}$ is a trade-off parameter. This scheme enables us to first pre-train our model as a standard VLM transformer model by mapping actions to text tokens ($\alpha = 0$), and then add additional action expert weights predicting continuous action tokens in a non-autoregressive fashion for fast inference in a post-training stage. We find that following this procedure, which is further explained below, leads to stable pre-training and excellent language following abilities of the VLA model. At inference time we then use standard autoregressive decoding for text tokens $\hat{\ell}$ followed by 10 denoising steps, conditioned on text tokens, to produce actions $\mathbf{a}_{t:t+H}$.

Archaeologist · close

What's new: mixture · unified HL/LL · VI · home-scale eval.

Lineage: $\pi_0 \rightarrow \text{FAST} \rightarrow \pi_{0.5} \rightarrow \text{KI} \rightarrow \text{RTC} \rightarrow \text{Hi Robot} \cdot \text{BEHAVIOR}'25 \rightarrow '26$

Q&A · Sep 30 · $\pi_{0.5}$ / 2504.16054

arXiv:2504.16054v1 [cs.LG] 22 Apr 2025



Abstract—In order for robots to be useful, they must perform practically relevant tasks in the real world, outside of the lab. While vision-language-action (VLA) models have demonstrated impressive results for end-to-end robot control, it remains an open question how far such models can generalize in the wild. We describe $\pi_{0.5}$, a new model based on π_0 that uses co-training on heterogeneous tasks to enable broad generalization. $\pi_{0.5}$ uses data from multiple robots, high-level semantic prediction, web data, and other sources to enable broadly generalizable real-world robotic manipulation. Our system uses a combination of co-training and hybrid multi-modal examples that combine image observations, language commands, object detections, semantic subtask prediction, and low-level actions. Our experiments show that this kind of knowledge transfer is essential for effective generalization, and we demonstrate for the first time that an end-to-end learning-enabled robotic system can perform long-

horizon and dexterous manipulation skills, such as cleaning a kitchen or bedroom, in entirely new homes.

I. INTRODUCTION

Stuff your eyes with wonder... See the world. It's more fantastic than any dream made or paid for in factories.

Ray Bradbury, *Fahrenheit 451*

Open-world generalization represents one of the biggest open problems in physical intelligence: embodied systems such as robotic arms, humanoids, and autonomous vehicles only truly become useful when they can leave the lab and handle the diverse situations and unexpected events that occur in the real world. Learning-based systems offer a path to en-